

Skill 2

```
manchodu@Srini:~$ pwd
/home/manchodu
manchodu@Srini:~$ ls
BSLAB      OSLAB      PSLAB      SDLAB      madhya.c    samantram.c
Chedu      OSLab      Parent     love.c.save nice.c.save
Chedu.c    OSSPLAB    Parent.c   madhya      samantram
manchodu@Srini:~$ mkdir LinuxLab
manchodu@Srini:~$ cd LinuxLab
manchodu@Srini:~/LinuxLab$ echo "Hello Linux" > file1.txt
manchodu@Srini:~/LinuxLab$ cat file1.txt
Hello Linux
manchodu@Srini:~/LinuxLab$ cp file1.txt file2.txt
manchodu@Srini:~/LinuxLab$ ls
file1.txt  file2.txt
manchodu@Srini:~/LinuxLab$ mv file2.txt renamed.txt
manchodu@Srini:~/LinuxLab$ ls
file1.txt  renamed.txt
manchodu@Srini:~/LinuxLab$ find . -name "renamed.txt"
./renamed.txt
manchodu@Srini:~/LinuxLab$ history
  1  cd
  2  cd ~
  3  pwd
  4  mkdir OSLab
  5  cd OSLab
  6  nano hello.c
  7  ls
  8  gcc hello.c -o hello
  9  ls
 10  ./hello
 11  ./hello5
 12  sudo apt update
 13  sudo apt install build-essential -y
 14  gcc --version
 15  cd ~/OSLab
 16  ls
 17  gcc hello.c -o hello
 18  ./hello
 19  mkdir OSLab
 20  cd OSLab
 21  nano process.c
 22  man sudo_root
 23  mkdir OSLab
 24  cd OSLab
 25  nano process.c
 26  .\process
 27  gcc process.c -o process
```

```
166 nano program6.c
167 gcc program6.c -o program6
168 cat program6.c
169 nano program6.1.c
170 gcc program6.1.c -o program6.1
171 ./program6.1
172 cat program6.c
173 nano program7.c
174 gcc program7.c -o program7
175 ./program7
176 cat program7.c
177 nano program8.c
178 gcc program8.c -o program8
179 ./program8
180 cat program8.c
181 pwd
182 ls
183 cd
184 cd Skill1
185 cd BSLAB
186 mkdir Skill1
187 rmdir Skill1
188 cp
189 cp Skill1
190 mv
191 pwd
192 ls
193 mkdir LinuxLab
194 cd LinuxLab
195 echo "Hello Linux" > file1.txt
196 cat file1.txt
197 cp file1.txt file2.txt
198 ls
199 mv file2.txt renamed.txt
200 ls
201 find . -name "renamed.txt"
202 history
manchodu@Srini:~/LinuxLab$ cd ..
manchodu@Srini:~$ rm -r LinuxLab
manchodu@Srini:~$ ls
BSLAB    OSLAB    PSLAB    SDLAB    madhya.c    samantram.c
Chedu    OSLab    Parent    love.c.save    nice.c.save
Chedu.c  OSSPLAB  Parent.c  madhya    samantram
manchodu@Srini:~$ |
```

```

manchodu@Srini:~/LinuxLab$ cd ..
manchodu@Srini:~$ rm -r LinuxLab
manchodu@Srini:~$ ls
BSLAB    OSLAB    PSLAB    SDLAB      madhya.c    samantram.c
Chedu    OSLab    Parent    love.c.save nice.c.save
Chedu.c  OSSPLAB  Parent.c  madhya     samantram
manchodu@Srini:~$ sudo apt update
[sudo] password for manchodu:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [897 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1174 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [198 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46.4 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1199 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [239 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [76.3 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1273 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [280 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [290 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [10.6 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1681 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [334 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1398 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [316 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [12.3 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:24 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5768 B]
Get:25 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [31.0 kB]
Get:26 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [12.6 kB]
Fetched 10.4 MB in 9s (1225 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
220 packages can be upgraded. Run 'apt list --upgradable' to see them.

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Get:25 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [31.0 kB]
Get:26 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [12.6 kB]
Fetched 10.4 MB in 9s (1225 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
220 packages can be upgraded. Run 'apt list --upgradable' to see them.
manchodu@Srini:~$ sudo apt install build-essential
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
build-essential is already the newest version (12.10ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 220 not upgraded.
manchodu@Srini:~$ gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
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This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

manchodu@Srini:~$ sudo apt install git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  git-man
Suggested packages:
  git-daemon-run | git-daemon-sysvinit git-doc git-email git-gui gitk
  gitweb git-cvs git-mediawiki git-svn
The following packages will be upgraded:
  git git-man
2 upgraded, 0 newly installed, 0 to remove and 218 not upgraded.
Need to get 4780 kB of archives.
After this operation, 1024 B of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 git-man all 1:2.43.0-1ubuntu7.3 [1100 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 git amd64 1:2.43.0-1ubuntu7.3 [3680 kB]
Fetched 4780 kB in 3s (1575 kB/s)
(Reading database ... 52290 files and directories currently installed.)
Preparing to unpack .../git-man_1%3a2.43.0-1ubuntu7.3_all.deb ...
Unpacking git-man (1:2.43.0-1ubuntu7.3) over (1:2.43.0-1ubuntu7.1) ...
Preparing to unpack .../git_1%3a2.43.0-1ubuntu7.3_amd64.deb ...
Unpacking git (1:2.43.0-1ubuntu7.3) over (1:2.43.0-1ubuntu7.1) ...
Setting up git-man (1:2.43.0-1ubuntu7.3) ...
Setting up git (1:2.43.0-1ubuntu7.3) ...
Processing triggers for man-db (2.12.0-4build2) ...

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Unpacking git-man (1:2.43.0-1ubuntu7.3) over (1:2.43.0-1ubuntu7.1) ...
Preparing to unpack ../git_1%3a2.43.0-1ubuntu7.3_amd64.deb ...
Unpacking git (1:2.43.0-1ubuntu7.3) over (1:2.43.0-1ubuntu7.1) ...
Setting up git-man (1:2.43.0-1ubuntu7.3) ...
Setting up git (1:2.43.0-1ubuntu7.3) ...
Processing triggers for man-db (2.12.0-4build2) ...
manchodu@Srini:~$ git --version
git version 2.43.0
manchodu@Srini:~$ mkdir newProject
manchodu@Srini:~$ cd newProject
manchodu@Srini:~/newProject$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/manchodu/newProject/.git/
manchodu@Srini:~/newProject$ mkdir src
manchodu@Srini:~/newProject$ mkdir include
manchodu@Srini:~/newProject$ mkdir bin
manchodu@Srini:~/newProject$ mkdir docs
manchodu@Srini:~/newProject$ ls
bin docs include src
manchodu@Srini:~/newProject$ echo "Hello World" > src/main.c
manchodu@Srini:~/newProject$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  src/

nothing added to commit but untracked files present (use "git add" to track)
manchodu@Srini:~/newProject$ git add .
manchodu@Srini:~/newProject$ git config --global user.name "Your Name"
manchodu@Srini:~/newProject$ git config --global user.email "you@gmail.com"
manchodu@Srini:~/newProject$ git commit -m "First Commit"
[master (root-commit) 511510d] First Commit
 1 file changed, 1 insertion(+)
 create mode 100644 src/main.c
manchodu@Srini:~/newProject$ ls

```

```

manchodu@Srini:~/newProject$ pstree
systemd--agetty
         |
         |--cron
         |--dbus-daemon
         |--init-systemd(Ub)
         |   |--SessionLeader--Relay(293)--bash--pstree
         |   |--init--{init}
         |   |--login--bash
         |   |--{init-systemd(Ub)}
         |--polkitd--3*[{polkitd}]
         |--rsyslogd--3*[{rsyslogd}]
         |--systemd--(sd-pam)
         |--systemd-journal
         |--systemd-logind
         |--systemd-resolve
         |--systemd-timesyn--{systemd-timesyn}
         |--systemd-udev
         |--unattended-upgr--{unattended-upgr}
manchodu@Srini:~/newProject$ ps
  PID TTY          TIME CMD
   293 pts/0      00:00:00 bash
  5133 pts/0      00:00:00 ps
manchodu@Srini:~/newProject$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1        0  0 03:30 ?           00:00:05 /sbin/init
root           2        1  0 03:30 hvc0       00:00:00 /init
root           6        2  0 03:30 hvc0       00:00:00 plan9 --control-socket 7
root          52        1  0 03:30 ?           00:00:01 /usr/lib/systemd/systemd
root          99        1  0 03:30 ?           00:00:00 /usr/lib/systemd/systemd
systemd+     111        1  0 03:30 ?           00:00:00 /usr/lib/systemd/systemd
systemd+     112        1  0 03:30 ?           00:00:00 /usr/lib/systemd/systemd
root        176        1  0 03:30 ?           00:00:00 /usr/sbin/cron -f -P
message+    177        1  0 03:30 ?           00:00:00 @dbus-daemon --system --
root        185        1  0 03:30 ?           00:00:00 /usr/lib/systemd/systemd
syslog      197        1  0 03:30 ?           00:00:00 /usr/sbin/rsyslogd -n -i
root        212        1  0 03:30 tty1        00:00:00 /sbin/agetty -o -p -- \u
root        215        1  0 03:30 ?           00:00:00 /usr/bin/python3 /usr/sh
root        286        2  0 03:30 ?           00:00:00 /init
root        288        286  0 03:30 ?           00:00:00 /init
manchodu    293        288  0 03:30 pts/0       00:00:00 -bash
root        297        2  0 03:30 pts/1       00:00:00 /bin/login -f
manchodu    345        1  0 03:30 ?           00:00:00 /usr/lib/systemd/systemd
manchodu    346       345  0 03:30 ?           00:00:00 (sd-pam)
manchodu    376       297  0 03:30 pts/1       00:00:00 -bash
polkitd     2299        1  0 03:35 ?           00:00:00 /usr/lib/polkit-1/polkit
manchodu    5146       293  0 03:44 pts/0       00:00:00 ps -ef
manchodu@Srini:~/newProject$ sudo apt install strace

```



```

manchodu@Srini:~/newProject$ ./exec
Before exec
bin    child    docs  exec.c  fork.c  include  process.c  shell.c  wait
bye.c  child.c  exec  fork    hello   process  shell      src      wait.c
manchodu@Srini:~/newProject$ cat bye.c
#include <stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
manchodu@Srini:~/newProject$ ./bye
-bash: ./bye: No such file or directory
manchodu@Srini:~/newProject$ ./hello
Hello Linux
manchodu@Srini:~/newProject$ |

```

```

manchodu@Srini:~/newProject$ ./bye
-bash: ./bye: No such file or directory
manchodu@Srini:~/newProject$ ./hello
Hello Linux
manchodu@Srini:~/newProject$ cat fork.c
#include<stdio.h>
#include<unistd.h>

int main()
{
    fork();

    printf("Hello\n");

    return 0;
}
manchodu@Srini:~/newProject$ ./fork
Hello
Hello
manchodu@Srini:~/newProject$ |

```

```
manchodu@Srini:~/newProject$ cat child.c
#include<stdio.h>
#include<unistd.h>

int main()
{
    pid_t pid;

    pid=fork();

    if(pid==0)
    {
        printf("Child PID=%d\n",getpid());
        printf("Parent PID=%d\n",getppid());
    }
    else
    {
        printf("Parent PID=%d\n",getpid());
    }

    return 0;
}
manchodu@Srini:~/newProject$ ./child
Parent PID=9523
Child PID=9524
Parent PID=8827
manchodu@Srini:~/newProject$ |
```

```
manchodu@Srini:~/newProject$ cat make.c
cat: make.c: No such file or directory
manchodu@Srini:~/newProject$ nano wait.c
manchodu@Srini:~/newProject$ cat wait.c
#include<stdio.h>
#include<unistd.h>
#include<sys/wait.h>

int main()
{
    if(fork()==0)
    {
        printf("Child\n");
    }
    else
    {
        wait(NULL);

        printf("Parent\n");
    }

    return 0;
}
manchodu@Srini:~/newProject$ ./wait
Child
Parent
manchodu@Srini:~/newProject$ |
```

```
manchodu@Srini:~/newProject$ cat process.c
```

```
#include<stdio.h>
#include<unistd.h>
```

```
int main()
{
    fork();
    fork();

    printf("PID=%d\n",getpid());

    return 0;
}
```

```
manchodu@Srini:~/newProject$ ./process
```

```
PID=9905
```

```
PID=9907
```

```
PID=9906
```

```
PID=9908
```

```
manchodu@Srini:~/newProject$ |
```